Domain-Adversarial Training Under Label Shift Can Silently Invert Out-of-Distribution Detection

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Abstract

Domain-adversarial training (DANN) is a standard remedy for site and device shortcuts in medical imaging, and deployed models increasingly rely on out-of-distribution (OOD) monitors to flag unfamiliar inputs. We show that the two can conflict silently. On skin-lesion classification with a lesion-disjoint ISIC 2019 split, gradient reversal toward smartphone images drives Mahalanobis OOD detection from AUROC 0.808 to 0.403 (95% CI $[0.380, 0.425]$), and to 0.402, 0.306 and 0.375 on three clinical datasets no part of training saw, while in-distribution balanced accuracy (-0.023, CI $[-0.063, +0.017]$) and calibration do not change. The inversion is caused by label shift between the domains the adversary aligns: changing only the class mix the adversary sees moves unseen-domain AUROC by 0.24–0.37, and the effect persists at matched domain leakage. Mechanistically, DANN pulls unfamiliar images toward the majority class; under label shift it reaches chance-level unweighted domain leakage by misaligning classes, while the domain stays decodable within each class. The consequence is clinical: the share of melanomas from unseen sources predicted benign nevus with confidence above 0.9 rises from 8% to 49–81%. Importance-weighted alignment recovers 32–45% of the lost AUROC at matched leakage; with the true target class priors detection returns to its pre-adaptation level, but no estimator we tried (BBSE, EM, 50 labelled target images) gets close enough to deploy. OOD monitoring should be re-validated after domain adaptation, and domain leakage should be measured within class.

Keywords: domain-adversarial training; out-of-distribution detection; label shift; calibration; skin lesion classification; deployment monitoring

1. Introduction

Skin lesion classifiers learn acquisition artefacts as readily as pathology: ink markings, rulers, dermatoscope type and contributing site all correlate with diagnosis in public datasets (Winkler et al., 2019; Bissoto et al., 2020; Daneshjou et al., 2022). The standard remedy is to make the representation invariant to the acquisition domain, most often by domain-adversarial training with a gradient reversal layer (Ganin et al., 2016; Bousmalis et al., 2016). A deployed model also needs a second safeguard: an out-of-distribution (OOD) monitor that flags inputs unlike its training data so that a clinician reviews them. This paper asks what invariance training does to that monitor.

In our setting it can turn the monitor upside down. As the adversarial weight grows, OOD detection does not merely weaken; it falls below chance, so that images from clinics the model has never seen score as *more* familiar than its own test data. Nothing in a standard evaluation reveals this: in-distribution accuracy and calibration stay flat, and the domain-leakage probe, the quantity the method is meant to reduce, improves. The cost

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falls on the patients a monitor exists to protect: under the same training, melanomas from unseen clinics are predicted benign nevus with confidence above 0.9 up to ten times more often, and the OOD monitor that should route them to a clinician ranks them as routine.

We trace the failure to label shift. ISIC 2019 is 50.8% nevus, the smartphone images the adversary aligns against are 12.3% nevus, and matching the marginal feature distributions of domains with different class priors forces the representation to misplace one domain's classes (Zhao et al., 2019; Tachet des Combes et al., 2020). We show the consequence for OOD monitoring with an intervention that changes only the class mix the adversary sees, and we locate the mechanism in the feature space.

Contributions.

1. **The failure.** Under a lesion-disjoint split, gradient reversal inverts Mahalanobis OOD detection on held-out smartphone images and on three clinical datasets that no branch of training saw, with flat in-distribution accuracy and calibration, under two training recipes whose accuracy is within 0.06 of a standard ERM baseline on the same split.

2. **The cause.** Setting the class mix seen by the adversary to that of the target removes the inversion on unseen domains (AUROC difference 0.24–0.37); a regression at matched domain leakage attributes the effect to label shift rather than to deeper invariance.

3. **The mechanism and a measurement lesson.** DANN pulls images that differ from the source along the domain direction toward the majority class. Under label shift, an unweighted domain probe reports chance-level leakage while a within-class probe still reads the domain: invariance should be measured within class.

4. **Consequence and mitigation.** Melanomas from unseen sources are called benign nevi with high confidence up to ten times more often. Importance weighting recovers part of the loss; with the true target class priors it recovers fully, which makes estimating target class priors the bottleneck for a deployable fix.

2. Related work

Shortcuts and invariance in dermatology. Ink markings shift melanoma predictions (Winkler et al., 2019), classifiers keep competitive ISIC accuracy with the lesion masked out (Bissoto et al., 2020), and performance degrades on darker skin (Daneshjou et al., 2022). Domain-adversarial training (Ganin et al., 2016) and domain separation networks (Bousmalis et al., 2016) are the usual remedies; our dual-encoder model is a domain separation network and is a vehicle for the intervention, not a contribution.

Invariance under label shift. When class priors differ between domains, matching marginal feature distributions cannot also keep the joint error small (Zhao et al., 2019); importance-weighted alignment (IWDAN, IWCDAN) corrects for this with estimated target class priors (Tachet des Combes et al., 2020), which can be estimated without target labels by black-box shift estimation (Lipton et al., 2018) or EM (Saerens et al., 2002). These results concern classification error. We study what the same misalignment does to OOD

monitoring, and show that a standard domain-leakage probe misreports invariance under label shift.

OOD detection after adaptation. Full-spectrum OOD detection (Yang et al., 2023; Zhang et al., 2023) asks that covariate-shifted in-distribution inputs be accepted; our smartphone and clinical images fall in that cell, and Section 5 argues that acceptance is safe only when the model also classifies them. Calibration degrades under shift (Ovadia et al., 2019) and after domain adaptation (Wang et al., 2020). Closest to our observation, Wu et al. report Mahalanobis and ODIN AUROC below 0.5 on rotated MNIST after adaptation with an MMD-based method (DAN) (Wu et al., 2023), as a baseline in a study of continually adaptive detection. We show the inversion on medical images, identify label shift as its cause, and characterise the mechanism.

3. Materials and methods

3.1. Data and splits

ISIC 2019 (Tschandl et al., 2018) (25,331 dermoscopic images, eight classes: MEL, NV, BCC, AK, BKL, DF, VASC, SCC) is the in-distribution source. We split it 60/20/20 into train, validation and test *grouped by* `lesion_id` (16,211 / 4,053 / 5,067 images), because 60% of test images in the image-level split share a lesion with training (Tran, 2027). **PAD-UFES-20** (Pacheco et al., 2020) (2,298 smartphone photographs, six of the eight classes) is the domain the adversary aligns against. Its labels are never used in any loss. We split it by patient into `pad_adv` (1,582 images, 961 patients), seen by the adversarial and context branches, and `pad_heldout` (716 images, 412 patients), never seen.

Unseen domains. Four clinical-photograph datasets enter no part of training: Fitzpatrick17k (Groh et al., 2021) (3,887 images obtained), MED-NODE (170 images), SD-198 (6,584 images) and SCIN (10,406 images, 5,033 cases) [PENDING: cite MED-NODE, SD-198, SCIN; licences: SCIN CC BY 4.0, MED-NODE none stated, SD-198 official source]. SCIN is mostly inflammatory conditions; only 0.6% of its images map to the ISIC classes, so we use it only as an OOD set for detection and report it separately; no label-based analysis is done on SCIN. Images are cropped to a lesion-centred box and resized to 224×224.

3.2. Model and training

A dual-encoder model with EfficientNet-B3 backbones produces a lesion latent $z_\ell \in \mathbb{R}^{16}$ (batch-normalised, z_ℓ^{norm}) used for diagnosis through a linear head, and a context latent $z_c \in \mathbb{R}^{64}$. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{ctx}}\mathcal{L}_{\text{ctx}} + \lambda_{\text{adv}}\,\mathcal{L}_{\text{adv}} + \lambda_{\text{orth}}\mathcal{L}_{\text{orth}}, \tag{1}$$

with $\mathcal{L}_{\text{cls}}$ on ISIC only, $\mathcal{L}_{\text{ctx}}$ a domain cross-entropy on z_c, $\mathcal{L}_{\text{adv}}$ a domain cross-entropy on z_ℓ^{norm} behind a gradient reversal layer, and $\mathcal{L}_{\text{orth}}$ a squared-cosine penalty between the latents. AdamW (learning rate 10^{-4}; adversary $\times 30$, classifier $\times 0.2$), 40 epochs, domain-balanced batches. We sweep $\lambda_{\text{adv}} \in \{0, 0.25, 0.5, 1, 2, 4, 8\}$ with everything else fixed (seeds 42–82 at $\lambda_{\text{adv}} \in \{0, 2\}$, 42–46 elsewhere). A second recipe (A4: $z_\ell \in \mathbb{R}^{64}$, classifier $\times 1$, class-weighted loss) serves as a robustness check.

Checkpoints. The sweep in Table 1 uses the best-validation checkpoint (selected epochs 6–39). All other experiments use the last checkpoint: validation balanced accuracy (eight classes, chance 0.125) has a lesion-clustered bootstrap SE of 0.023–0.026, and the best-validation and last checkpoints differ by less than one SE, so best-validation selection picks early epochs by noise. Both are reported [PENDING: Table 1 at last checkpoint (t1last)].

Competence. In-distribution balanced accuracy at $\lambda_{\text{adv}} = 0$ is 0.496 (main recipe) and 0.520 (A4), 0.04–0.06 below a standard single-encoder ERM on the same split (0.556 [PENDING: ERM reference at 5 seeds, with SE]) and four times chance (0.125). Lesion-disjoint eight-class ISIC 2019 is hard; the image-level split inflates the same model's balanced accuracy by 0.20 (Tran, 2027).

3.3. Interventions on label shift

Class mix seen by the adversary. Keeping the classification loss on all of ISIC, we resample only the ISIC images fed to the adversarial branch so that their class mix is ISIC's own (100%, the default), halfway to PAD's (50%), or PAD's (0%, no label shift between the aligned domains), at $\lambda_{\text{adv}} \in \{0.5, 2\}$ with five seeds (three at 50%).

Importance-weighted alignment. IWDAN and IWCDAN (Tachet des Combes et al., 2020) reweight the source in the adversarial loss by estimated target class priors; we compare BBSE estimates (Lipton et al., 2018), EM with temperature-scaled probabilities (Saerens et al., 2002), priors from 50 labelled `pad_adv` images (a few-label setting), and the true `pad_adv` priors (oracle, not deployable). CDAN (Long et al., 2018) is a class-conditional adversarial baseline.

3.4. Measurements

OOD detection. Class-conditional Mahalanobis with shared covariance (Lee et al., 2018) (primary), k-NN (Sun et al., 2022), maximum cosine to a class centroid, maximum softmax probability (Hendrycks and Gimpel, 2017) and energy (Liu et al., 2020), all fit on ISIC train. AUROC treats OOD as the positive class, so 0.5 is chance and values below 0.5 mean OOD images score as more familiar than ID test images. The five scores are functions of the same latent, so they are not independent evidence.

Inversion. We call a cell inverted when the cluster-bootstrap 95% CI of the seed-mean Mahalanobis AUROC lies below 0.5 and at least three of five seeds' own intervals do [PENDING: confirm against code: 3/5 or 5/5; threshold for 3-seed cells (2/3 or 3/3)]. Bootstrap draws (2,000) resample ISIC test by lesion and OOD sets by patient (`pad_heldout`), near-duplicate cluster (Fitzpatrick17k, SD-198) or case (SCIN); MED-NODE has no grouping and is resampled by image, so its intervals may be too narrow [PENDING: confirm SD-198 uses P13 clusters in every table]. Runs with in-distribution balanced accuracy below 0.25 are excluded as degraded and reported.

Domain leakage. Balanced accuracy (chance 0.5) of a logistic probe for the source dataset on frozen z_ℓ^{norm}. The probe used in earlier work scores ISIC test against all of PAD, including images the adversary trained on. We therefore also report probes on held-out images only (linear, one-hidden-layer MLP, k-NN), a class-balanced probe that reweights both domains to a common class mix, and a within-class probe averaged over classes.

Table 1: The intervention, under the lesion-disjoint ISIC split. Mean ± s.d. across five seeds at every λ_{adv}. Leakage is balanced accuracy (chance 0.5); cross-domain accuracy is balanced over the six classes shared with PAD-UFES (chance 0.167). Arrows mark the direction a practitioner would call favourable. The image-level split, which inflates in-distribution accuracy by 0.20, and the two non-adversarial controls are in Appendix A.1.

λ_{adv}	n	Leakage ↓	ID bal. acc ↑	ID ECE	OOD ECE	OOD AUROC	Fitz. AUROC	X-dom bal. acc
0	5	0.927 ± 0.018	0.496 ± 0.016	0.181 ± 0.026	0.219 ± 0.046	$\mathbf{0.808 \pm 0.027}$	0.717 ± 0.034	0.275 ± 0.023
0.25	5	0.614 ± 0.092	0.487 ± 0.005	0.141 ± 0.076	0.561 ± 0.221	0.525 ± 0.121	0.496 ± 0.167	0.289 ± 0.049
0.5	5	0.581 ± 0.031	0.501 ± 0.012	0.191 ± 0.008	0.710 ± 0.017	$\mathbf{0.426 \pm 0.008}$	0.411 ± 0.047	0.259 ± 0.025
1	5	0.563 ± 0.045	0.495 ± 0.011	0.176 ± 0.016	0.730 ± 0.015	$\mathbf{0.387 \pm 0.016}$	0.375 ± 0.042	0.229 ± 0.027
2	5	0.555 ± 0.046	0.473 ± 0.023	0.181 ± 0.027	0.717 ± 0.022	$\mathbf{0.403 \pm 0.028}$	0.402 ± 0.018	0.254 ± 0.015
4	5	0.636 ± 0.076	0.498 ± 0.012	0.161 ± 0.050	0.696 ± 0.032	$\mathbf{0.433 \pm 0.020}$	0.376 ± 0.037	0.258 ± 0.010
8	5	0.663 ± 0.068	0.489 ± 0.019	0.161 ± 0.029	0.710 ± 0.027	$\mathbf{0.454 \pm 0.044}$	0.412 ± 0.051	0.254 ± 0.022

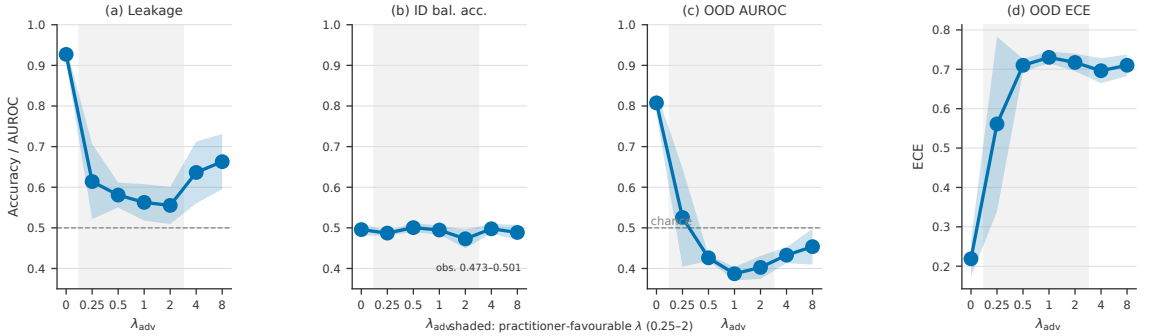


Figure 1: **The intervention, four views, one axis.** Leakage falls to its chance floor and in-distribution balanced accuracy does not degrade — both favourable. Over the same range OOD AUROC collapses through chance and inverts, and out-of-domain calibration error triples. A developer monitoring the first two panels would see nothing wrong.

Calibration and diagnosis. Expected calibration error (15 bins) on ISIC test and on `pad_heldout` (six shared classes), and zero-shot balanced accuracy on `pad_heldout`. Out-of-domain ECE is close to mean confidence minus accuracy here, so these are views of one phenomenon.

4. Results

4.1. The intervention

Table 1 reports every measurement as a function of λ_{adv}, the only manipulated variable. Figure 1 shows the four quantities that matter most on a shared axis.

Read across a single row and nothing is wrong. Read down the columns and two of them improve while two collapse.

4.2. Leakage falls, in-distribution metrics do not warn, and invariance is never completed

Domain leakage falls from 0.927 at $\lambda_{\text{adv}} = 0$ to a minimum of 0.555 at $\lambda_{\text{adv}} = 2$, against a chance floor of 0.5. **It never reaches that floor**, and it is not monotone: at the two largest weights it rises again, to 0.636 and 0.663. So the intended effect of the method is obtained in part and then partly undone, and a practitioner reading this column alone would stop increasing λ_{adv} somewhere near $\lambda_{\text{adv}} = 2$.

There is no in-distribution cost over that range. Balanced accuracy on the ISIC test split stays between 0.473 and 0.501 at every λ_{adv} — the paired per-seed difference between $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 2$ is -0.023 with a 95% CI of $[-0.063, +0.017]$, so we claim no change in either direction — and in-distribution expected calibration error is flat at ≈ 0.18. A practitioner tracking accuracy, in-distribution calibration and the leakage probe would conclude that the method had worked as advertised.

4.3. Out-of-distribution detection collapses, and inverts

Mahalanobis AUROC against the held-out PAD partition falls from 0.808 at $\lambda_{\text{adv}} = 0$ to 0.426 at $\lambda_{\text{adv}} = 0.5$, reaches its minimum of 0.387 at $\lambda_{\text{adv}} = 1$, and remains between 0.387 and 0.454 at every larger weight. Three features matter more than the size of that drop.

The descent is complete by $\lambda_{\text{adv}} = 0.5$, and $\lambda_{\text{adv}} = 0.25$ sits on the edge. At $\lambda_{\text{adv}} = 0.25$ the value is 0.525 ± 0.121: the seed spread straddles chance, and **we claim no inversion there**. From $\lambda_{\text{adv}} = 0.5$ onward every weight is below chance, with no ordering a practitioner could tune against. So detection collapses between $\lambda_{\text{adv}} = 0$ and $\lambda_{\text{adv}} = 0.5$ and is inverted thereafter; there is no operating point at which invariance pressure has been applied and detection survives.

It goes below chance. At $\lambda_{\text{adv}} = 2$ the detector ranks images from a clinic the model has never seen as *more* in-distribution than its own test split. A patient-clustered bootstrap puts the 95% interval at $[0.380, 0.425]$ with all five seeds' individual intervals below 0.5, and k-NN (0.427) and cosine (0.420) agree. On the image-level split, where the robustness checks below were run, the corresponding value is 0.427, $[0.414, 0.439]$, 5/5 seeds, and it is not a tail artefact: removing the pooled 90th, 95th or 99th percentile leaves AUROC at 0.412, 0.416 and 0.424 against 0.427 for the full set, and the medians themselves invert (ISIC 13.8 versus `pad_heldout` 10.0).

The inversion does not require domain invariance, and does not track it. Leakage never reaches its floor: the minimum over the whole sweep is 0.555 against a chance level of 0.5, so the encoder remains domain-decodable at every weight. More tellingly, leakage *rises* again at the largest weights — 0.636 at $\lambda_{\text{adv}} = 4$ and 0.663 at $\lambda_{\text{adv}} = 8$, against 0.555 at $\lambda_{\text{adv}} = 2$ — while AUROC stays below chance throughout (0.433 and 0.454). Invariance is therefore neither achieved nor monotone in λ_{adv}, yet the inversion persists. This is the sharpest form of the dissociation developed in Section 4.5: whatever gradient reversal does to the representation, the leakage probe does not measure it.

All five rules move together — MSP 0.970 $\rightarrow$ 0.437, energy 0.983 $\rightarrow$ 0.441, cosine 0.901 $\rightarrow$ 0.435, k-NN 0.941 $\rightarrow$ 0.424 on the image-level split, stable across $k \in \{10, 50, 200\}$ — but as noted in the methods they are five views of one 16-dimensional representation;

the independent replications are the backbone (Section 4.5) and the single-encoder model (Section 4.7).

Three artefact explanations fail, all tested on the image-level split. *Covariance estimation:* Ledoit–Wolf shrinkage leaves $\lambda_{\text{adv}} = 2$ at 0.427, and across float32 and float64, with and without shrinkage, at one, four and eight BLAS threads, the largest AUROC deviation is 9.4×10^{-4}; effective dimensionality barely moves (participation ratio 4.879 → 4.500 while leakage falls 0.914 → 0.553 in the same three runs). *An over-strong adversary:* the adversary runs at 30× the base learning rate, and retraining at 10× and 100× keeps the inversion — 0.418 ± 0.011 at 10× with all three seeds below 0.5, against 0.432 ± 0.024 at 30× and 0.482 ± 0.020 at 100× (one seed at 0.503) — so weakening it tenfold deepens the inversion rather than removing it. *The adversary's own domain:* on Fitzpatrick17k, absent from every training branch, AUROC falls 0.717 → 0.402 under the leak-free split and 0.764 → 0.391 under the image-level one (95% CI $[0.386, 0.396]$, five of five seeds below 0.5; bootstrapped over images, so likely anti-conservative). For context a ResNet-50 baseline reaches 0.994 on those images and the EfficientNet-B3 single-encoder control 0.649; the controlled comparison is the $\lambda_{\text{adv}} = 0$ row, which differs from $\lambda_{\text{adv}} = 2$ only in the adversarial weight.

4.4. The model becomes confidently wrong on unseen domains

Confidence rises as competence falls. At $\lambda_{\text{adv}} = 0$ mean maximum-softmax confidence is 0.915 on ISIC test and 0.489 on `pad_heldout`; by $\lambda_{\text{adv}} = 2$ the ordering has inverted, 0.930 on a clinic the model has never seen against 0.910 on its own test set, while cross-domain balanced accuracy never improves at any λ_{adv} (0.249–0.291 against a floor of 0.167) and predictions collapse toward the benign majority class, with **actinic keratosis recall falling** 0.245 → 0.016. Out-of-domain calibration error triples, 0.247 → 0.746, while in-distribution ECE stays flat at ≈ 0.10, so no in-distribution measurement reveals it. That adapted models are miscalibrated on the target domain is known (Wang et al., 2020); what the sweep adds is the dose-dependence and the invisibility, and we stress that confidence, accuracy and calibration here are three views of one phenomenon rather than three results (Appendix A.2).

4.5. Leakage does not mediate the collapse

The natural hypothesis is that removing domain information is what breaks detection. Four results rule it out, the last of them outside dermatology.

First, the two quantities are not even monotonically related. Leakage bottoms out at 0.555 and *rises* to 0.636 and 0.663 at $\lambda_{\text{adv}} = 4$ and 8, while AUROC stays below chance throughout (0.433, 0.454). Detection is at its worst (0.387) at $\lambda_{\text{adv}} = 1$, where leakage is 0.563 — higher than at $\lambda_{\text{adv}} = 2$. The damage neither waits for invariance nor follows it.

Second, at the pre-projection backbone, leakage is essentially flat across the whole sweep (0.982 → 0.941) while backbone Mahalanobis AUROC falls 0.749 → 0.475. Domain information remains linearly decodable there, and detection collapses anyway. Because the backbone is a different representation from the 16-dimensional latent, this is the independent replication that the five-detector agreement does not provide.

Third, and most directly, two training objectives traverse the same leakage range with opposite consequences (Section 4.6). Leakage alone therefore does not determine the outcome.

Fourth, in histopathology the two quantities come apart almost completely. In a two-hospital Camelyon17 setting (three seeds, hospital-disjoint slides held out for detection), raising λ_{adv} from 0 to 1 moves all five detectors by 0.12 to 0.22 — Mahalanobis 0.651 → 0.534, cosine 0.785 → 0.589, k-NN 0.794 → 0.648, MSP 0.798 → 0.589, free energy 0.794 → 0.575 — while the hospital probe moves by 0.007 ($0.959 \pm 0.007 \rightarrow 0.952 \pm 0.001$), which is inside its own seed spread at $\lambda_{\text{adv}} = 0$, and in-distribution accuracy is unchanged at 0.995. The adversary reached chance here (final cross-entropy 0.692 against $\ln 2 = 0.693$), so the optimisation pressure was applied; the encoder simply did not become invariant under it. Detection degraded regardless. **This is the clearest dissociation we obtained: the outcome moved roughly twenty-five times as far as the supposed mediator.**

Two qualifications. The detectors fall but remain above chance (0.53–0.65), so this is degradation, not the collapse to chance seen in dermatology, and nothing here bears on the inversion. And these runs come from an experiment that failed its own pre-registered gate — we had required the probe to fall by at least 0.10 before treating the setting as a test of invariance, and it fell by 0.007. That gate failure is precisely what makes the runs informative here: they are a case where invariance was *not* obtained and detection degraded anyway. We report them for this claim only, and the pre-registered verdict on the original question stands as negative (Section 6).

We accordingly report leakage as a measured quantity and never as the explanatory variable, and plot all results against λ_{adv} rather than against leakage.

4.6. The failure is specific to gradient reversal

Adversarial training is one route to invariance among several. Table 2 sweeps four alternatives on the single-encoder backbone: deep CORAL (Sun and Saenko, 2016), maximum mean discrepancy (Long et al., 2015), invariant risk minimisation (Arjovsky et al., 2019) and group distributionally robust optimisation (Sagawa et al., 2020).

Only MMD moves leakage appreciably, and it does so at a price gradient reversal never pays. Escalating the MMD weight from 1 to 10,000 buys 0.13 of leakage (0.967 → 0.836, floor 0.5) at a cost of 0.44 of in-distribution balanced accuracy (0.779 → 0.337), and leaves every detector well above chance (Mahalanobis 0.665, k-NN 0.738, energy 0.694 at the highest weight). IRM reaches the extreme of the same pattern: at weight 100 its balanced accuracy collapses to 0.167 while leakage remains at 0.950. CORAL does not induce invariance at all; at weight 100 it makes the representation *more* domain-separable (leakage 0.998). GroupDRO leaves leakage unmoved at every weight tested.

The contrast is cleanest when task health is held constant rather than leakage, since MMD only reaches comparable leakage by destroying the classifier:

Both arms' detectors are scored on `pad_heldout`; the leakage probe in this comparison uses `pad_full`. At $\lambda_{\text{adv}} = 0.25$ gradient reversal reduces leakage 0.970 → 0.763 **at no cost in accuracy** (0.845) and puts all five detectors at chance (0.51–0.55). MMD at weight 100 reaches leakage 0.900 and collapses nothing: Mahalanobis 0.811 → 0.724 and every detector stays at or above 0.66. Pushed to weight 10,000 it reaches leakage 0.836

— still above anything gradient reversal needs — but only by destroying the task, with in-distribution balanced accuracy at 0.337, and even then the detectors hold at 0.63–0.74.

The calibration signatures differ in the same direction. As MMD degrades, its out-of-domain ECE *falls* (0.344 → 0.142): the model becomes less confident as it becomes less accurate, which is the appropriate response. Under gradient reversal out-of-domain ECE triples while accuracy falls. The two objectives move leakage the same way and fail in opposite manners — one degrades honestly, the other is confidently wrong.

We do not claim MMD would be safe at the invariance level gradient reversal reaches; it never got there. What the data establish is that at the leakage levels MMD does reach, it shows none of the failure, while gradient reversal at the same levels has already collapsed.

4.7. It is not specific to the factorized architecture

All results above use a dual-encoder model. Repeating the sweep with plain single-encoder domain-adversarial training — one backbone, one linear classifier, a gradient reversal layer and a domain head, with no context branch and no orthogonality term — reproduces the cliff: every detector collapses at $\lambda_{\text{adv}} = 0.25$ and by $\lambda_{\text{adv}} = 4$ four of the five sit below chance, while in-distribution accuracy stays flat (Appendix A.3, Table 3). **The invisibility, the collapse and the inversion are therefore properties of domain-adversarial training rather than of the factorized architecture.**

A second backbone leaves the question open. Repeating the single-encoder sweep with a ResNet-50 backbone did not settle whether the inversion is specific to EfficientNet-B3, because comparable invariance was never reached. At $\lambda_{\text{adv}} = 2$ the adversary never left chance — best-epoch cross-entropy 0.686 to 0.688 against a threshold of 0.673 — so those three runs are null experiments and we draw no conclusion from them. At $\lambda_{\text{adv}} = 0.5$, where the adversary did learn (best-epoch cross-entropy 0.41 to 0.55), leakage fell only from 0.965 to 0.939 against the 0.742 that EfficientNet-B3 reaches on the same partition, so invariance was not obtained and the backbone question is untested rather than answered in the negative.

On that run the detectors disagree, and we report all of them rather than the one that suits each point. Over the same $\lambda_{\text{adv}} = 0 \rightarrow 0.5$ step, k-NN falls 0.962 → 0.563 and cosine 0.922 → 0.543, while Mahalanobis moves only to 0.814 and never approaches chance. Three detectors that move together on EfficientNet-B3 separate here. We draw two things from this, and no more. Taking the detectors that do move, the step is the sharpest non-mediation evidence in the paper: the outcome shifts by 0.40 while leakage, the supposed mediator, shifts by 0.026 (Section 4.5). Taking Mahalanobis, the headline detector everywhere else in this paper, nothing happens. **The claim that all five scoring rules collapse together therefore holds on EfficientNet-B3 and does not hold here**, which is an additional reason the backbone question is open.

One difference is informative. The single-encoder model plateaus at leakage 0.76–0.81 and never reaches the 0.571 the dual-encoder attains on that split; more adversarial pressure buys no further invariance. The dual-encoder has somewhere to put the discarded information — a domain probe on its context branch reads > 0.9999 — while the single-encoder must destroy it. The architecture that achieves deeper invariance also suffers the larger

collapse (0.427 versus 0.509 at $\lambda_{\text{adv}} = 2$), which is the dose–response relationship seen from a second direction.

4.8. The collapse generalises to a same-modality site shift; the inversion does not

Every result above contrasts dermoscopy with clinical photography, so the construct we call domain is confounded with imaging modality. ISIC 2019 contains a contrast that is not: its `lesion_id` field records the contributing site, with 12,413 images from Hospital Clínic Barcelona (`BCN`) and 10,015 from the HAM10000 collection (`HAM`) — two hospitals, two dermatoscopes, two patient populations, one modality. The two sites are separable from frozen ERM features at balanced accuracy 0.962 ± 0.007, comparable to the 0.970 that ISIC↔PAD reaches, and not for trivial reasons: masking the dermatoscope vignette costs only 0.019, hand-crafted colour statistics reach 0.901 against a 0.876 ceiling on the original pair, and a label-only probe reaches 0.668 (Appendix A.5).

The collapse reproduces. With `BCN` in distribution and a lesion-disjoint 30% `HAM` partition held out, leakage falls $0.963 \pm 0.006 \rightarrow 0.821 \pm 0.030$ at $\lambda_{\text{adv}} = 0.25$ and the detectors with headroom fall through chance: k-NN $0.658 \pm 0.041 \rightarrow 0.486 \pm 0.009$, cosine $0.576 \pm 0.040 \rightarrow 0.477 \pm 0.006$ (3 seeds). Unlike the two multi-domain settings in the limitations, the manipulation took.

The inversion does not. Reversing the roles restores detector headroom — Mahalanobis reads 0.846 ± 0.003 at $\lambda_{\text{adv}} = 0$, essentially the 0.843 of the dermatology sweep — and adversarial training then reduces leakage from 0.95 to 0.77 and drives Mahalanobis to 0.683 ± 0.015 at $\lambda_{\text{adv}} = 1$, where the patient-clustered bootstrap gives [0.666, 0.699] with all three seeds' intervals entirely above 0.5. Detection degrades substantially and stops above chance.

Reading. Baseline detectability is the same (0.846 versus 0.843) and the achieved invariance is the same as the single-encoder dermatology run (0.77 versus 0.763), yet the cross-modality contrast reaches 0.427 while the same-modality one stops at 0.683. Neither a shortage of headroom nor a shortfall in achieved invariance accounts for the gap, so we separate two claims the earlier sections report together. The *collapse* — detection falling to chance as invariance is obtained, invisibly to in-distribution metrics — reproduces between two hospitals. The *inversion* does not: in every setting where we observe it the two domains differ in imaging modality. Whether that reflects a threshold in the size of the shift, something specific to modality gaps, or a property of these datasets, we cannot say.

4.9. What the leak-free split costs, and what it does not change

All results above use an ISIC split grouped by `lesion_id`, because 3,041 of 5,067 test images (60.0%) share a lesion with training or validation under the image-level split that most published work on this dataset uses. Removing that overlap costs 0.20 of in-distribution balanced accuracy at $\lambda_{\text{adv}} = 0$ — 0.692 image-level against 0.496 leak-free — which is an order of magnitude larger than any effect we attribute to λ_{adv}, and consistent with the companion analysis of group overlap in deployment monitoring (Tran, 2027). **It changes no conclusion here**: every pattern in Table 1 reproduces under the image-level split

(Appendix A.1), and the class distribution of the two test sets differs by at most one image in any class, so the gap is leakage rather than composition.

4.10. The inversion runs through the majority-class centroid

Nevus is 50.8% of the in-distribution set. We pre-registered three criteria for the hypothesis that gradient reversal moves out-of-domain images into the nevus region of the latent space, and all three hold.

(i) Out-of-domain images migrate to the nevus cell. The fraction of `pad_heldout` images whose nearest class centroid is nevus rises from 0.332 at $\lambda_{\text{adv}} = 0$ to 0.601 at $\lambda_{\text{adv}} = 2$ ($\Delta = 0.268$ against $2\,\text{SE} = 0.055$), while the same fraction for in-distribution images is flat ($0.528 \to 0.531$). It tracks the diagnostic shift: across all 27 runs the correlation between this fraction and the number of `pad_heldout` images predicted nevus ($105 \to 424$ of 716) is $\rho = 0.982$.

(ii) The inversion is confined to those images. Split by nearest centroid, AUROC on the nevus-nearest subset falls $0.807 \to 0.282$ while the remainder stays above chance throughout ($0.861 \to 0.644$).

(iii) They move, and the reference does not. Against a class-matched reference — in-distribution images that are themselves nevus — the median squared Mahalanobis distance from out-of-domain images to the nevus centroid falls from 49.8 to 17.5, while the reference distance is unchanged ($7.5 \to 7.4$).

The pattern replicates on the pre-projection backbone, a different representation of 1536 dimensions (nevus-nearest fraction $0.484 \to 0.603$, nevus-nearest AUROC $0.726 \to 0.330$), so it is not a property of the 16-dimensional projection.

What the mechanism does not say. Out-of-domain images never become more typical than real nevi: the distance ratio stays above 1 at every weight ($6.63 \to 2.37$). The inversion is a property of the *pooled* AUROC, which mixes classes. Pooled AUROC equals the class-share-weighted mean of the per-class AUROCs, since AUROC averages over the in-distribution samples — an identity, not a result, which we use only to locate the effect. **The content is the two per-class values: 0.559 against nevus and 0.296 weighted over the other five.** Out-of-domain images are close to indistinguishable from nevi, whose centroid they are drawn toward, and inverted against everything else; because the majority class is half the data, the pooled score crosses chance. With all eight classes present the identity holds to machine precision at every weight, both out-of-domain sets and both in-distribution sets (largest residual 2.2×10^{-16}); restricted to the six PAD-shared classes the discrepancy reaches 0.010, the weight of the two omitted classes. It also explains why per-class and pooled AUROC disagree at $\lambda_{\text{adv}} = 0$ (Section 5).

This accounts exactly for a result we had reported and could not previously explain. Reweighting the in-distribution set to PAD's class mix — 12.3% nevus against ISIC's 50.8% — *strengthens* the inversion to 0.259 ± 0.022 (Table 4), because thinning nevus from the reference makes images sitting in the nevus region look comparatively more typical. Reweighting changes only the weights w_c; the model, every image's score and the out-of-domain set are untouched, so no per-class AUROC moves, and evaluating $\sum_c w_c \text{AUROC}_c$ under PAD's mix reproduces the measurement to 8×10^{-7}. A two-group reduction does not, giving 0.306,

which is why the magnitude appeared not to follow from the mechanism until the full form was used.

One consequence is worth stating with its scope. **For the pooled score, with the model and out-of-domain set fixed, the inversion is deeper the *less* the in-distribution set is dominated by its majority class**: the two-group reduction $0.559p + 0.296(1-p)$ for a nevus share p, with the non-nevus classes at ISIC proportions, returns above chance only at $p = 0.776$. That is a statement about one metric under one model, not a rule about deployments.

4.11. What the phenomenon is not

Table 4 in the appendix lists six candidate accounts — memorisation of the adversarial images, class composition, domain specificity, dimensional collapse, latent compression and tail asymmetry — each tested against a prediction fixed before its test, and each refuted, together with a pre-registered synthetic reproduction that failed to reproduce the inversion. We report them because they are the reason the account in Section 4.10 is credible: it is what survived after the alternatives were eliminated, rather than the first story we tried.

5. Discussion

An inversion, not a degradation. The central result is not that out-of-distribution detection weakens under adversarial invariance training but that the ordering reverses. Degradation would be unsurprising — a representation stripped of domain information should be worse at recognising a domain, and chance is what that argument predicts, which is what a single-encoder model reaches at $\lambda_{\text{adv}} = 0.25$ (0.525). Going *past* chance is not predicted by it, and is what Section 4.10 accounts for. The results then compose into one failure sequence: a model trained with invariance to acquisition domain, as recommended, is deployed at a new site, where it increasingly predicts the benign majority class — actinic keratosis recall $0.245 \rightarrow 0.016$ — with higher confidence than on its own test data, while the monitor that should flag those images ranks them as more routine than familiar ones and every metric the developer watches moves favourably.

The failure belongs to gradient reversal, not to invariance. Reducing leakage is not sufficient to produce it, and in this sweep leakage is not even achieved: it bottoms out at 0.555 against a floor of 0.5 and rises again at the largest weights while detection stays inverted. MMD reduces leakage along the same axis and collapses nothing at the levels it reaches with the classifier intact, where gradient reversal has already inverted. The two also fail in opposite manners: pushed harder, MMD becomes less accurate *and* less confident and its out-of-domain calibration error falls, while under gradient reversal accuracy falls as confidence rises and calibration error triples. One degrades honestly, the other is confidently wrong. Wald et al. (Wald et al., 2021) showed that enforcing multi-domain calibration *produces* invariance; the converse fails here, which suggests that where invariance is the goal, calibration-based or moment-matching objectives may be preferable because they leave the out-of-domain uncertainty signal intact. We offer that as a hypothesis: MMD never reached the invariance level gradient reversal reaches without destroying the task,

and we tested one adversarial formulation, so whether CDAN or adversarial discriminative adaptation behaves the same way is not established here.

The danger zone is where the method succeeds. Risk scales with *achieved* invariance, not with the nominal weight or the presence of an adversarial term. Where the adversary never learned, leakage stayed at 0.94 across four orders of magnitude of λ_{adv} and the safety failure did not appear either. A weak adversary produces neither debiasing nor harm — so many deployed configurations may be harmless, and a configuration's safety cannot be read off its hyperparameters.

Tolerance without generalisation. Full-spectrum OOD detection (Yang et al., 2023) prescribes that covariate-shifted in-distribution inputs be accepted rather than flagged. Our models achieve that prescription and are unsafe because of it: they accept out-of-domain images more readily than their own test data while classifying them at 0.254 balanced accuracy against a floor of 0.167. Acceptance and competence are separate properties, and current protocols evaluate them separately or not at all. Any claim of tolerance to covariate shift should carry cross-domain accuracy on the same inputs, since tolerating inputs a model cannot classify is a failure mode rather than a success.

Recommendation. Report OOD detection and out-of-domain calibration **as a function of achieved invariance** rather than at a single operating point; report cross-domain accuracy alongside any claim of tolerance to covariate shift; and state, when a separate dataset serves as the OOD set, that this measures covariate shift rather than semantic novelty. Practitioners tuning λ_{adv} already train at several values, so this changes what is reported rather than what is run. It converts an invisible failure into a visible number.

Monitor outside the adversarial path. Since the collapse is a property of the representation that received adversarial gradient, one mitigation is to score novelty on a representation that did not. A frozen ImageNet ResNet-50 gives Mahalanobis AUROC 0.988 on Fitzpatrick17k and 0.833 on `pad_heldout`, against the trained model's 0.764 and 0.843 at $\lambda_{\text{adv}} = 0$ and 0.391 and 0.427 at $\lambda_{\text{adv}} = 2$. It never inverts, but **we had pre-registered** 0.90 **in all four cells and `pad_heldout` does not reach it**, where the frozen encoder merely matches the untrained model rather than improving on it. A partial mitigation, not a fix; retreating to an earlier layer of the *same* encoder does not work at all (Section 4.5).

The inversion is deepened, not created, in the minority classes. Per-class AUROC, which unlike the pooled value does not depend on the ID class mix, is 0.14–0.39 on `pad_heldout` and 0.10–0.34 on Fitzpatrick17k at $\lambda_{\text{adv}} = 2$ for every non-nevus class. Two are already at or below chance beforehand: on `pad_heldout` at $\lambda_{\text{adv}} = 0$, actinic keratosis is 0.499 and squamous cell carcinoma 0.444. Gradient reversal deepens an inversion that already exists for the rarest and most clinically consequential classes, and the pooled AUROC at $\lambda_{\text{adv}} = 0$ conceals this because nevus is half the in-distribution set.

The mechanism, and what remains open. Gradient reversal removes the acquisition signal by moving out-of-domain images into the region the majority class occupies (Section 4.10) — an efficient way to satisfy the adversary, since the densest part of the latent space is where domain information is hardest to recover, and simultaneously the worst outcome for a distance-based novelty score, which reads density as familiarity. The two

objectives are not merely in tension: invariance is obtained by the move that defeats detection. What remains open is why gradient reversal takes this route and moment matching does not, why the synthetic reproduction failed, and whether the majority class is special or merely the densest available — a dataset with two comparably sized dominant classes would separate the last of these.

6. Limitations

Absolute cross-domain performance is poor for every model here. Plain accuracy on `pad_heldout` is 0.245 for the factorized model and 0.391 for the baseline against a 0.385 majority rate, so no model here is deployable. Our claim concerns the *effect of* λ_{adv}, and the deployment pathway above is a mechanism of harm rather than a description of a fielded system.

The inversion is observed only across imaging modality. ISIC is dermoscopy while PAD-UFES and Fitzpatrick17k are clinical photography, so the main contrast confounds site with modality and Fitzpatrick17k does not resolve it. The same-modality contrast within ISIC (Section 4.8) separates the claims: the collapse reproduces between two hospitals, the inversion does not. We cannot say whether that reflects the size of the shift, modality gaps specifically, or these datasets.

Three attempts to replicate outside dermatology returned no result, for the same underlying reason in different guises: the manipulation did not take, so there was nothing to measure. On three-hospital Camelyon17 the adversary head never trained; on iWildCam it trained but the encoder never became invariant; and on Camelyon17 restricted to *two* hospitals — the obvious diagnosis, matching the dermatology domain count, with the gradient path verified first — the adversary reached chance ($\ln 2$) while a post-hoc probe still read the hospital at 0.92–0.95, a fall of at most 0.037 against the 0.10 we had pre-committed. The pre-registered stopping rule returns a negative and we stop. **Domain count is not the barrier**, and we cannot say what is; the pattern — an adversary driven to chance while the information it was asked to remove stays linearly decodable — is a known failure mode of adversarial removal (Elazar and Goldberg, 2018). **We report no inversion outside dermatology**; a single-seed sub-chance reading there did not replicate (0.534 ± 0.050 over three seeds). What those runs do support is the dissociation in Section 4.5. Details in Appendix A.4.

Generality beyond dermatology remains untested. **Backbone coverage.** Comparable invariance was not reached with a ResNet-50 backbone, so whether the inversion extends beyond EfficientNet-B3 is untested. Three of the nine runs failed their adversary gate outright and are reported as null experiments rather than negative results. **The objective comparison is bounded.** MMD never reached the invariance level gradient reversal attains; it could only approach it by destroying the classifier. We therefore cannot say what moment-matching would do at matched invariance, only that at the leakage levels it reaches, gradient reversal has already inverted and MMD has not.

Detector non-independence. The five scoring rules are functions of one 16-dimensional latent; the backbone analysis and the single-encoder replication carry the independent evidence. **Calibration arithmetic.** Out-of-domain ECE is close to mean confidence minus accuracy here, so those three results are one phenomenon viewed three ways. **Seeds.** Five

at every λ_{adv} on the leak-free split, and three at two interior points of the image-level sweep in the appendix; effect sizes are large relative to seed variance, but we claim no formal test across seven points.

Splits. The main sweep uses the image-level ISIC split, in which 3,041 of 5,067 test images (60.0%) share a `lesion_id` with training or validation. This inflates in-distribution balanced accuracy by 0.20 (0.496 → 0.692 at $\lambda_{\text{adv}} = 0$), so the absolute values in Table 1 are optimistic by that amount; Section 4.9 reports the lesion-disjoint replication at five seeds, under which every conclusion is unchanged. PAD splits are patient-level, disjoint by patient and by lesion. **Third-domain coverage.** 3,887 of 16,577 Fitzpatrick17k images were obtained; the set is web-scraped under mixed copyright, used for evaluation only, and not redistributed.

7. Conclusion

Under domain-adversarial training, out-of-distribution detection does not merely weaken. It reverses: a model asked which inputs are unfamiliar answers with its own test data and ranks a new clinic's images as the more routine of the two. The reversal appears at the smallest adversarial weight that achieves invariance, holds on a dataset the adversary never saw, reproduces without the factorized architecture, and does not appear under moment-matching objectives that reduce leakage along the same axis.

It is invisible to everything a developer monitors. In-distribution accuracy holds, in-distribution calibration is unchanged and the domain leakage probe improves exactly as intended — under an image-level and a lesion-disjoint ISIC split alike — while confidence on unseen domains overtakes confidence on familiar data and recall for a premalignant class falls to 0.016.

The cause is identifiable: invariance is obtained by moving out-of-domain images into the majority class's region of the latent space, which is exactly the move that makes a density-based novelty score call them familiar. Acting on it does not require the explanation. Reporting out-of-distribution detection, out-of-domain calibration and cross-domain accuracy as functions of achieved invariance costs almost nothing and makes the failure visible.

Data and code availability

https://github.com/TranDucVinh-20225538/CSG-SKin. ISIC 2019 CC BY-NC 4.0; PAD-UFES-20 CC BY 4.0; Fitzpatrick17k annotations CC BY-NC-SA 3.0 with images used for evaluation only and not redistributed.

Declaration of competing interest

The authors declare no competing interests.

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Figure 6: same-modality site shift

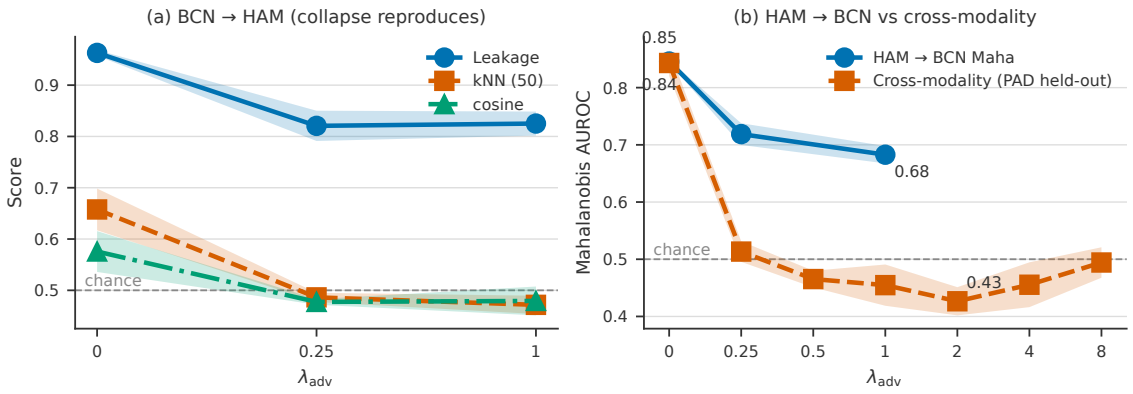


Figure 2: **A same-modality site shift separates the two claims.** (a) Barcelona to HAM10000: leakage falls and the detectors with headroom cross chance. (b) With the roles reversed to restore headroom, Mahalanobis falls from a baseline matching the cross-modality sweep but stops above chance, while the cross-modality sweep continues below it.

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Appendix A. Supporting tables and figures

A.1. The image-level split

Table 5 is the same sweep under the image-level ISIC split used by most published work on this dataset, in which 60.0% of test images share a `lesion_id` with training. Every pattern is the same; the in-distribution accuracies are about 0.20 higher throughout.

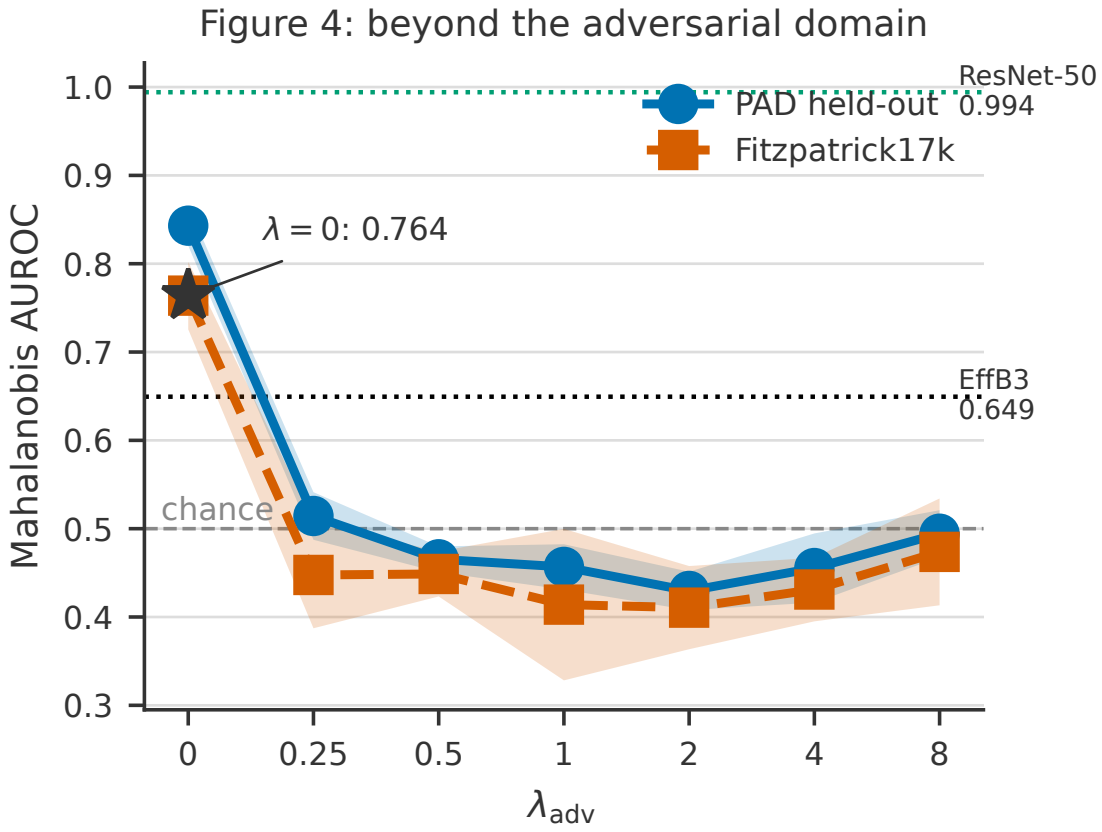


Figure 3: **The collapse is not confined to the adversarial domain.** AUROC against λ_{adv} for the held-out PAD partition and for Fitzpatrick17k, which is absent from every training branch. Horizontal references give the ResNet-50 baseline and the EfficientNet-B3 control on Fitzpatrick; the controlled comparison is the $\lambda_{\text{adv}} = 0$ point of the sweep itself.

A.2. Confidence, accuracy and calibration on unseen domains

At $\lambda_{\text{adv}} = 0$ the model is appropriately hesitant on unfamiliar data: mean maximum-softmax confidence is 0.915 on the ISIC test split and 0.489 on `pad_heldout`. By $\lambda_{\text{adv}} = 0.25$ that gap has closed (0.913 versus 0.898), and by $\lambda_{\text{adv}} = 2$ it has inverted (0.930 versus 0.910).

Cross-domain diagnostic accuracy does not improve at any λ_{adv}: balanced accuracy spans 0.249–0.291 against a six-class floor of 0.167, no setting exceeds the 0.291 obtained at $\lambda_{\text{adv}} = 0$, and every setting is below the 0.298 of the non-adversarial EfficientNet-B3 control. Plain accuracy falls 0.245 → 0.184, beneath the 0.385 majority-class rate throughout. Macro AUC is roughly preserved (0.605 → 0.581), so discrimination survives while the decision rule does not.

Predictions collapse toward the benign majority class: nevus predictions rise from 113 to 374 of 716 images, and actinic keratosis recall falls from 0.245 to 0.016 (49/202 → 3/202

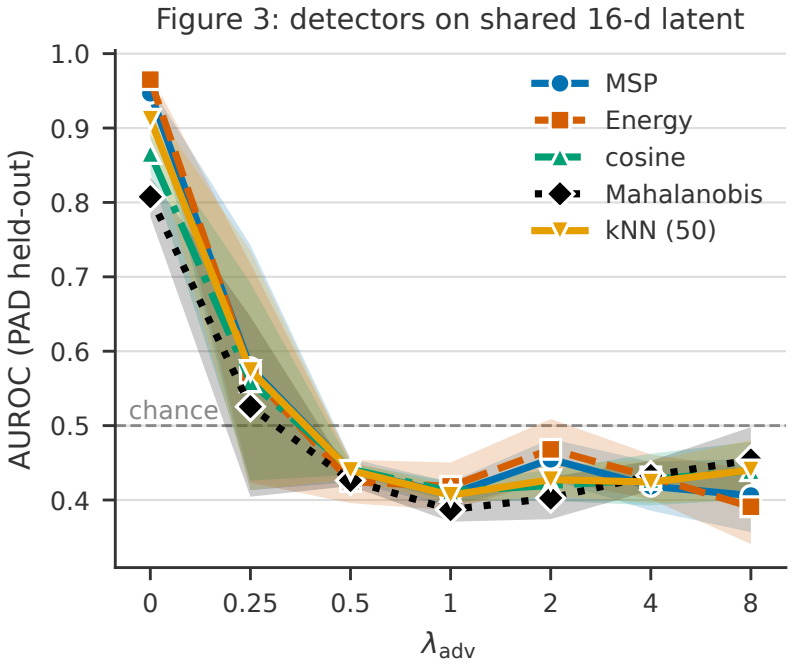


Figure 4: **Every scoring rule moves together.** The five detectors against λ_{adv} on the held-out PAD partition, mean ± 1 s.d. All five are functions of the same 16-dimensional latent and are therefore not independent evidence; Figure 6 carries the replication on a different representation.

images, mean over five seeds). Squamous cell carcinoma moves the same way (0.088 → 0.021, 6/66 → 1/66) but the count is too small to carry a claim on its own, and melanoma ($n = 9$) supports none.

In this setting out-of-domain ECE is arithmetically close to mean confidence minus accuracy, which is why the confidence, accuracy and calibration results are not independent corroboration of each other.

A.3. Single-encoder replication

Plain single-encoder domain-adversarial training on the same data and splits. Every detector collapses at $\lambda_{adv} = 0.25$ — k-NN 0.944 → 0.528, energy 0.911 → 0.507, cosine 0.884 → 0.509, MSP 0.868 → 0.515 — and by $\lambda_{adv} = 4$ four of the five sit below chance, cosine most clearly at 0.454 ± 0.014. In-distribution accuracy is flat throughout (0.845 → 0.832) and cross-domain accuracy falls from 0.390 to a plateau near 0.32.

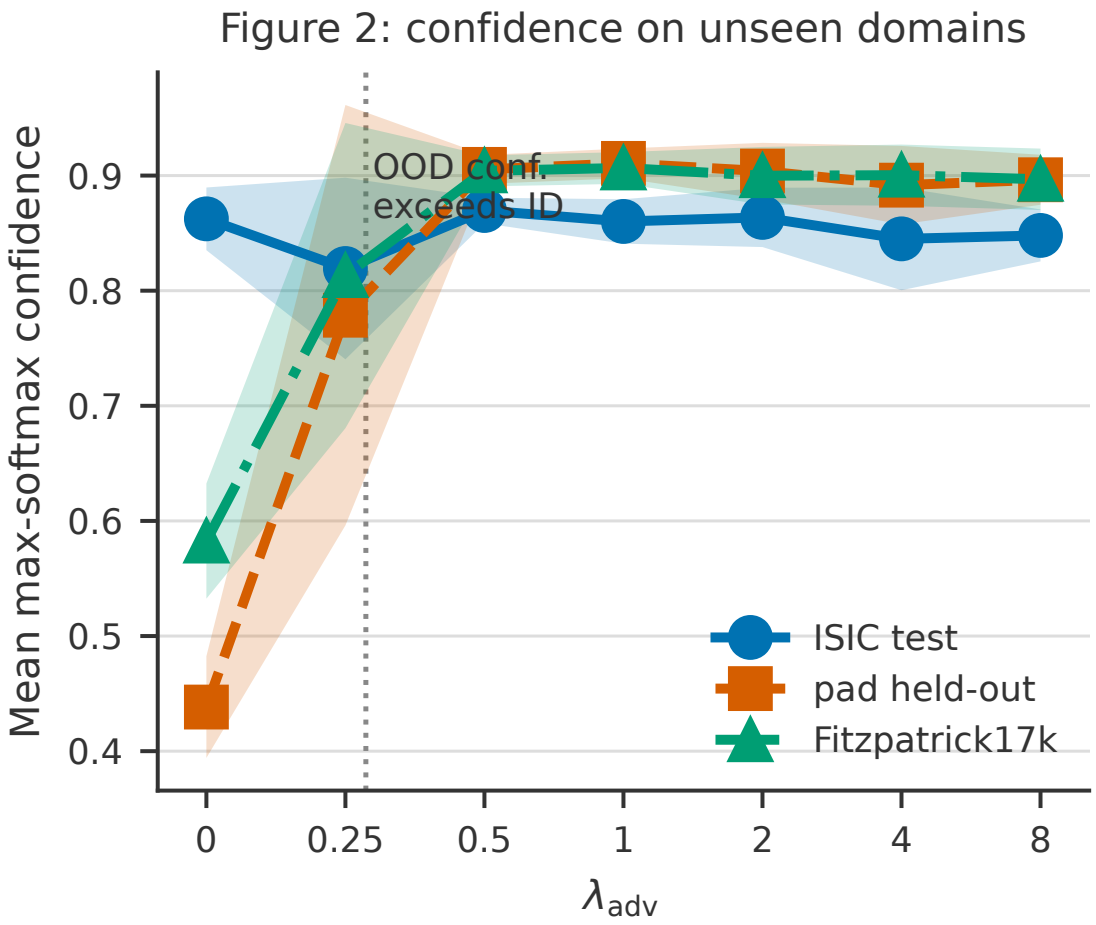


Figure 5: **Confidence on unseen domains overtakes confidence on familiar data.** Mean maximum-softmax probability against λ_{adv} for the ISIC test split, the held-out PAD partition and Fitzpatrick17k. The crossing occurs by $\lambda_{\text{adv}} \approx 0.5$. λ spacing is categorical.

A.4. External replication attempts

Camelyon17, three hospitals. The adversary head never trained: cross-entropy sat at $\ln 3$ from the first epoch at every λ_{adv} while a post-hoc probe on the same features read 0.94, despite a gradient graph verified correct (head cosine +1.000, encoder −1.000).

iWildCam. The adversary did train — best-epoch cross-entropy 3.13 against $\ln 243 = 5.49$ — but the encoder never became invariant: the binary train-versus-OOD site probe moved only from 0.911 to 0.891 across $\lambda_{\text{adv}} \in [0, 10]$ against a floor of 0.5, and no detector collapsed.

Camelyon17, two hospitals. Pre-committed in `PRECOMMIT_CAMELYON2.json` before running. The adversary reached chance (final cross-entropy 0.692 against $\ln 2 = 0.693$,

Figure 5: leakage flat, OOD score falls (shared y-range)

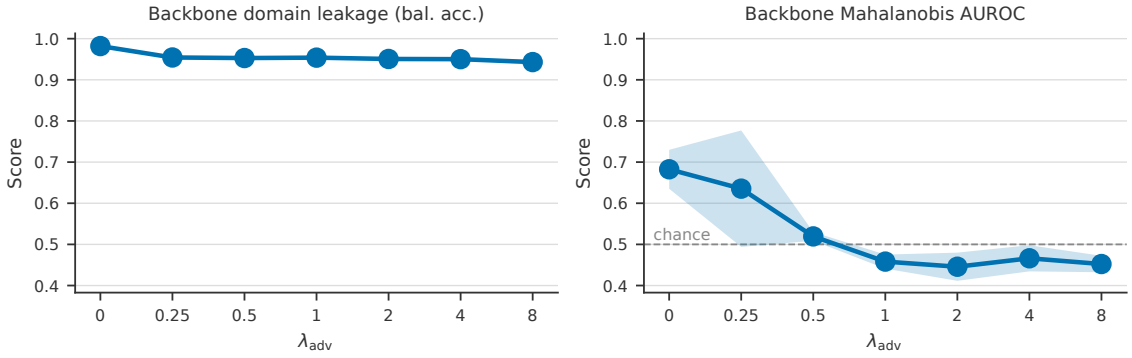


Figure 6: **Leakage does not mediate the collapse.** At the pre-projection backbone the domain probe stays flat across the sweep while Mahalanobis AUROC falls. Domain information remains linearly decodable and detection degrades regardless.

Table 2: Invariance objectives other than gradient reversal, single-encoder backbone, 3 seeds per cell. Leakage is balanced accuracy, floor 0.5. **IRM and GroupDRO use PAD labels** through their environment and group risks; the remaining objectives do not, so their cross-domain numbers are not comparable and the two regimes are kept in separate blocks.

Objective	Weight	Leakage	Maha. AUROC	ID bal. acc	OOD ECE
No access to PAD labels					
ERM[§]	—	0.973	0.838	0.761	—
CORAL	100	0.998	0.999	0.777	—
MMD	1	0.967	0.813	0.779	—
MMD	100	0.900	0.716	0.730	—
MMD	3000	0.893	0.748	0.382	0.142
MMD	10000	0.836	0.665	0.337	0.267
GRL	$\lambda_{adv} = 0.25$	**0.763**	**0.525**	**0.845**[†]	—
Uses PAD labels — not comparable on cross-domain accuracy					
IRM	100	0.950	0.729	0.167	—
GroupDRO	10	0.976	0.868	0.783	—

[†] plain accuracy for the GRL row; see Table 3.

[§] scored on `pad_full` with a uniform optimiser and an ISIC-only loss; the λ_{adv}=0 row of Table 3 uses `pad_heldout`, grouped learning rates and `ignore_index`. They are not the same condition. [PENDING: harmonise or relabel.]

from 0.11 at $\lambda_{adv} = 0$) and in-distribution accuracy held at 0.995, but the hospital probe fell only $0.959 \pm 0.007 \rightarrow 0.952 \pm 0.001$, inside its own seed spread, so the leakage gate failed and the verdict is negative. The exploratory seeds added afterwards do not reopen it: Mahalanobis on the held-out hospital reads 0.534 ± 0.050 at $\lambda_{adv} = 1$ over seeds 42–44,

Table 3: Single-encoder DANN on the same data and splits. ID accuracy is plain accuracy; leakage is balanced accuracy, floor 0.5. All five detectors collapse at the smallest non-zero weight, and four of five fall below chance by $\lambda_{\text{adv}} = 4$.

λ_{adv}	n	Leakage	Maha.	k-NN	MSP	cosine	Energy	ID acc	X-dom
0	5	0.970	0.700	**0.944**	0.868	0.884	**0.911**	0.845	0.390
0.25	3	**0.763**	0.525	**0.528**	0.515	0.509	**0.507**	0.845	0.333
0.5	3	0.795	0.510	0.517	0.501	0.493	0.507	0.844	0.328
1	3	0.795	0.548	0.521	0.462	0.482	0.448	0.840	0.334
2	5	0.807	0.509	0.516	0.479	0.468	0.470	0.839	0.325
4	3	0.798	0.502	0.486	0.471	0.454	0.449	0.835	0.320
8	3	0.787	0.520	0.532	0.519	0.490	0.500	0.832	0.321

Table 4: Candidate explanations tested and refuted. Each was a prediction specified before its test.

Explanation	Test	Result
Memorisation of the adversarial images	Hold PAD out of the adversary at patient level	`pad_heldout` 0.427 vs `pad_adv` 0.407; unchanged
Class composition	Restrict ID to the six shared classes; reweight ID to the OOD class mix	0.432 vs 0.427 on `pad_heldout`; reweighting *strengthens* the inversion to 0.259
Domain specificity	Evaluate on Fitzpatrick17k, never in $\mathcal{L}_{\text{adv}}$	0.391 at $\lambda_{\text{adv}} = 2$ against 0.764 at $\lambda_{\text{adv}} = 0$, same model and data
Dimensional collapse	Participation ratio and eigenspectrum across λ_{adv}	4.879 $\rightarrow$ 4.500 while leakage 0.914 $\rightarrow$ 0.553 (same three runs); decoupled
Latent compression	Latent variance and distance to centroid across λ_{adv}	OOD is already 4.5$\times$ tighter at $\lambda_{\text{adv}} = 0$; the adversary *decompresses* it
Tail asymmetry	Recompute AUROC after truncating the pooled upper tail	0.412–0.424 versus 0.427; medians invert
Synthetic reproduction	Pre-registered toy model with heavy-tailed ID structure, grid fixed before fitting	No inversion (λ_{adv}=0: 0.693; λ_{adv}=2: 0.786)

above chance, so the single-seed 0.478 did not replicate. These runs appear in Section 4.5 for the non-mediation claim only, where the five detectors move 0.12–0.22 while the probe moves 0.007.

A.5. Same-modality site shift: probe validation and the `HAM` direction

The sites are separable, and not for trivial reasons. A logistic probe on frozen ERM features distinguishes `BCN` from `HAM` at balanced accuracy 0.962 ± 0.007 against a floor of 0.5. Masking the dermatoscope vignette reduces this only to 0.943 ± 0.005, so the border is not the signal. Hand-crafted colour and intensity statistics reach 0.901 at best — close to the 0.876 ceiling the same features reach on ISIC$\leftrightarrow$PAD, so this pair is no more low-

Table 5: The intervention under the image-level ISIC split, with the two non-adversarial controls. Five seeds at $\lambda_{\text{adv}} \in \{0, 0.25, 1, 2, 8\}$, three at $\lambda_{\text{adv}} \in \{0.5, 4\}$.

λ_{adv}	n	Leakage ↓	ID bal. acc ↑	ID ECE	OOD ECE	OOD AUROC	Fitz. AUROC	X-dom bal. acc
0	5	0.915 ± 0.021	0.692	0.107	0.247	0.843 ± 0.023	0.764 ± 0.038	0.291 ± 0.016
0.25	5	0.571 ± 0.027	0.692	0.100	0.694	0.509 ± 0.023	0.460 ± 0.057	0.295 ± 0.028
0.5	3	0.555 ± 0.029	0.701	0.096	0.724	0.465 ± 0.015	0.449 ± 0.025	0.270 ± 0.034
1	5	0.591 ± 0.035	0.695	0.097	0.735	0.457 ± 0.026	0.414 ± 0.086	0.258 ± 0.012
2	5	0.584 ± 0.027	0.707	0.099	0.746	0.427 ± 0.025	0.391 ± 0.068	0.249 ± 0.030
4	3	0.548 ± 0.035	0.686	0.100	0.726	0.455 ± 0.039	0.431 ± 0.036	0.276 ± 0.025
8	5	0.625 ± 0.045	0.679	0.099	0.713	0.494 ± 0.027	0.474 ± 0.060	0.269 ± 0.023
EffNet-B3 control		0.802	0.683	0.096	—	0.726 ± 0.021	0.649	0.298 ± 0.010
ResNet-50 baseline		0.979	0.655	0.101	—	0.868 ± 0.019	0.994	0.391

level than the original contrast — while sharpness statistics sit at chance (0.532). A probe given only the ground-truth class label reaches 0.668, reflecting the two sites' different class distributions; the feature probe is far above that reference.

Mahalanobis is uninformative in the `BCN`-in-distribution direction. It reads 0.379 at $\lambda_{\text{adv}} = 0$, already below chance before any adversarial training. Reweighting the in-distribution set to the out-of-domain class mix raises it to 0.487, so most of that baseline inversion is class composition — `HAM` is two-thirds nevus and therefore concentrated near a single well-populated centroid — with a residual we attribute to HAM10000's more standardised acquisition and correspondingly lower feature variance. This is why the reversed direction, which restores headroom, carries the scope claim in Section 4.8. In that direction the class-reweighted variant falls from 0.74 to 0.54, and Mahalanobis reads 0.719 ± 0.019 at $\lambda_{\text{adv}} = 0.25$.

A.6. A domain monitor is cheap, and semantic novelty is unaffected

A linear domain head on a frozen ImageNet ResNet-50 that has never seen this data reaches AUROC 0.998; on the trained baseline, 0.997; on the trained EfficientNet-B3 control, 0.991. Detecting the acquisition domain is easy from almost any representation, so retaining a monitor is not in itself difficult. What distinguishes the dual-encoder's context branch is *concentration*: a single principal component carries 84.5% of its variance and already separates the domains (AUROC > 0.9999, two discordant pairs in 11.6M), against 0.55–0.70 at $k = 1$ for every other representation tested.

The damage is specific to covariate shift. With the domain held constant and diagnostic classes withheld from training instead, both branches land between 0.6 and 0.8 and the ordering depends on which class is withheld (DF and VASC: lesion 0.648, context 0.641, baseline 0.687; SCC: lesion 0.806, context 0.607, baseline 0.684). No near/far structure organises the difference (Pearson -0.12 against a centroid-similarity index). We report this as a null result.